

MTH202 FOR FINAL TERM A

LECTURE 23 TO 45

PROVIDE BY ORANGE MONKEY TEAM

Lecture No.23

Mathematical Induction

What is PRINCIPLE OF MATHEMATICAL INDUCTION?

Let $P(n)$ be a propositional function defined for all positive integers n . $P(n)$ is true for every positive integer n if

1. Basis Step:

The proposition $P(1)$ is true.

2. Inductive Step:

If $P(k)$ is true then $P(k+1)$ is true for all integers $k \geq 1$. i.e. $\forall k, P(k) \rightarrow P(k+1)$

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Lecture No.24

Mathematical Induction for

Divisibility

Type of MATHEMATICAL INDUCTION?

1. DIVISIBILITY PROBLEMS
2. INEQUALITY PROBLEMS

DIVISIBILITY:

Let n and d be integers and $d \neq 0$. Then n is divisible by d or d divides n written as $d \mid n$. iff $n = d \cdot k$ for some integer k . alternatively,

We say that n is a multiple of d

D is a divisor of n

D is a factor of n.

Thus $d \mid n \Leftrightarrow \exists$ an integer k such that $n = d \cdot k$

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Lecture No.25 Methods of proof

What are methods of proof?

A proof is a logical argument that tries to show that a statement is true. There are many different ways to go about proving something,

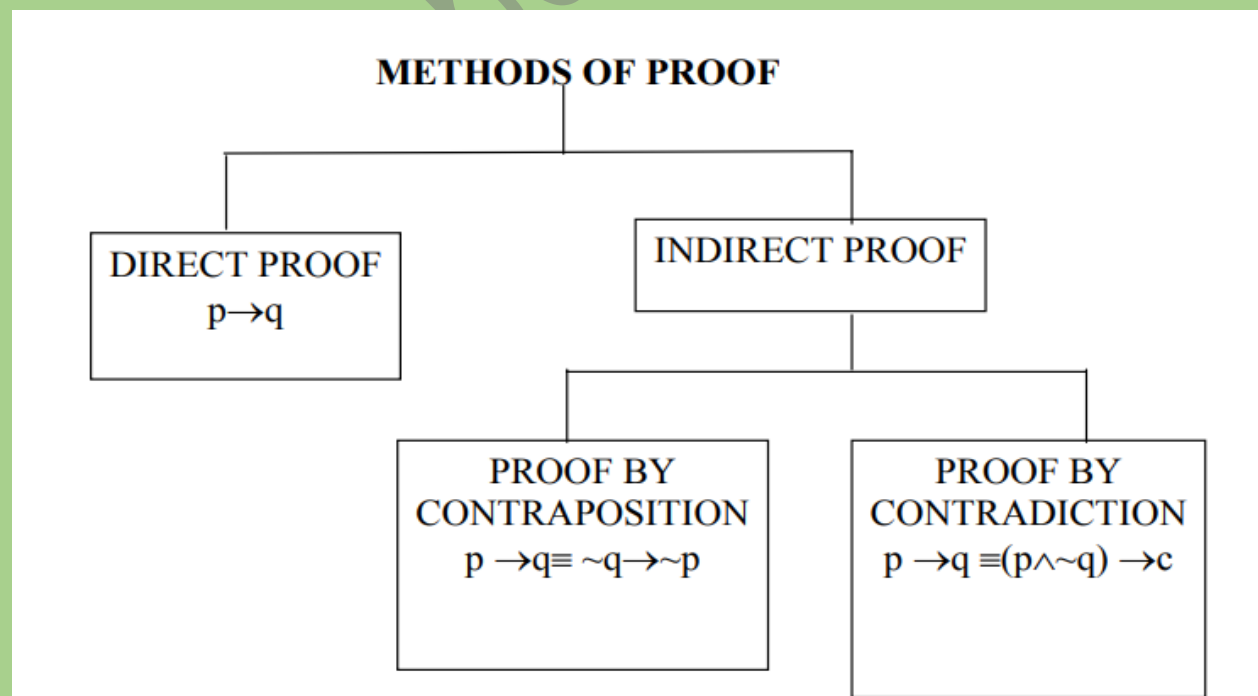
We'll discuss 2 methods:

1. Direct proof,
2. INDIRECT PROOF

Type of INDIRECT PROOF

1. Proof by contradiction,
2. Proof by induction.

We'll talk about what each of these proofs are, when and how they're used.



What is DIRECT PROOF?

The implication $p \rightarrow q$ can be proved by showing that if p is true, the q must also be true. This shows that the combination p true and q false never occurs. A proof of this kind is called a direct proof.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

SOME BASICS:

QUIZ IMPORTANT IN 3quiz and final

1. An integer n is even if, and only if, $n = 2k$ for some integer k .
2. An integer n is odd if, and only if, $n = 2k + 1$ for some integer k .
3. An integer n is prime if, and only if, $n > 1$ and for all positive integers r and s , if $n = r \cdot s$, then $r = 1$ or $s = 1$.
4. An integer $n > 1$ is composite if, and only if, $n = r \cdot s$ for some positive integers r and s with $r \neq 1$ and $s \neq 1$.
5. A real number r is rational if, and only if, a/b for some integers a and b with $b \neq 0$.
6. If n and d are integers and $d \neq 0$, then d divides n , written $d \mid n$ if, and only if, $n = d \cdot k$ for some integers k .
7. An integer n is called a perfect square if, and only if, $n = k^2$ for some integer k .

Question

imp

Assume that m and n are particular integers. Justify your answer to each of the following:

- 1) Is $4 + 6m + n$ even?
- 2) Is $8 + 5mn$ odd?

Solution:

- 1) Yes: $4m + 6n = 2 \cdot (2m + 3n)$ [by definition of even] $(2m + 3n)$ is an integer because $2, 3, m, n$ are integers and products and sums of integers are integers.
- 2) Yes: $8mn + 5 = 2 \cdot (4mn + 2) + 1$ [by definition of odd] $(4mn + 2)$ is an integer because $4, 2, m, n$ are integers and products and sums of integers are integers.

Question

imp

Prove that if x divides y and y divides z then x divides z .

Solution:

By our assumptions, and the definition of divisibility, there are natural numbers k_1 and k_2 such that $y = xk_1$ and $z = yk_2$

Consequently,

$$z = yk_2 = z = xk_1 k_2$$

Let $k = k_1 k_2$ now k is a natural number and $z = xk$. So by the definition of divisibility, x divides z .

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Lecture No.26 Proof by Contradiction

PROOF BY CONTRADICTION:

Many theorems in mathematics are conditional statements ($p \rightarrow q$). Now the negation of the implication $p \rightarrow q$ is

$$\sim (p \rightarrow q) \equiv \sim (\sim p \vee q)$$

$$\sim (p \rightarrow q) \equiv \sim (\sim p) \wedge (\sim q) \quad \text{DeMorgan's Law}$$

$$\sim (p \rightarrow q) \equiv p \wedge \sim q$$

Clearly if the implication is true, then its negation must be false, i.e., leads to a contradiction.

Hence $\sim (p \rightarrow q) \equiv (p \wedge \sim q) \rightarrow c$, where c is a contradiction

THEOREM:

short question and quiz

The sum of any rational number and any irrational number is irrational.

$$r = a / b \dots\dots\dots(1)$$

$$r + s = c / d \dots\dots\dots(2)$$

for some integers a, b, c and d

with $b \neq 0$ and $d \neq 0$.

Using (1) in (2), we get

$$a / b + s = c / d$$

$$s = (bc - ad) / bd$$

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Lecture No.27 Algorithm

THE DIVISION ALGORITHM:

Given any integer n and a positive integer d , there exist unique integers q and r such that $n = d \cdot q + r$ and $0 \leq r < d$.

$$n = 54,$$

$$d = 4 \quad 54 = 4 \cdot 13 + 2;$$

Hence

$$q = 13, r = 2$$

THEOREM:

Let a while loop with guard G be given, together with pre- and post-conditions that are predicates in the algorithm variables. Also let a predicate $I(n)$, called the loop invariant, be given. If the following four properties are true, then the loop is correct with respect to its pre- and post-conditions.

I. Basis Property: The pre-condition for the loop implies that $I(0)$ is true before the first iteration of the loop.

II. Inductive property: If the guard G and the loop invariant $I(k)$ are both true for an integer $k \geq 0$ before an iteration of the loop, then $I(k + 1)$ is true after iteration of the loop.

III. Eventual Falsity of Guard: After a finite number of iterations of the loop, the guard becomes false.

IV. Correctness of the Post-Condition: If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.

Lecture No.28

Lecture No.28 Division algorithm

HAND CALCULATION OF gcd:

Use the Euclidean algorithm to find $\gcd(330, 156)$

SOLUTION:

$$\begin{array}{r} 2 \\ 156 \overline{) 330} \\ \underline{312} \\ 18 \end{array} \qquad \begin{array}{r} 8 \\ 18 \overline{) 156} \\ \underline{144} \\ 12 \end{array}$$
$$\begin{array}{r} 1 \\ 12 \overline{) 18} \\ \underline{12} \\ 6 \end{array} \qquad \begin{array}{r} 2 \\ 6 \overline{) 12} \\ \underline{12} \\ 0 \end{array}$$

EXAMPLE:

Use the Euclidean algorithm to find $\gcd(330, 156)$

Solution:

1. Divide 330 by 156:

$$\text{This gives } 330 = 156 \cdot 2 + 18$$

2. Divide 156 by 18:

$$\text{This gives } 156 = 18 \cdot 8 + 12$$

3. Divide 18 by 12:

$$\text{This gives } 18 = 12 \cdot 1 + 6$$

4. Divide 12 by 6:

$$\text{This gives } 12 = 6 \cdot 2 + 0$$

Hence $\gcd(330, 156) = 6$.

LEMMA:

If a and b are any integers with $b \neq 0$ and q and r are nonnegative integers such that $a = q \cdot b + r$

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Lecture No.29 Combinatory

THE SUM RULE:

If one event can occur in n_1 ways, a second event can occur in n_2 (different) ways, then the total number of ways in which exactly one of the events (i.e., first or second) can occur is $n_1 + n_2$.

SUM RULE IN TERMS OF SETS:

If $A_1, A_2, \dots, A_m$ are finite disjoint sets, then the number of elements in the union of these sets is the sum of the number of elements in them. If $n(A_i)$ denotes the number of elements in set A_i for $i = 1, 2, \dots, m$, then

$$N(A_1 \cup A_2 \cup \dots \cup A_m) = n(A_1) + n(A_2) + \dots + n(A_m) \text{ where } A_i \cap A_j = \phi \text{ if } i \neq j$$

GENERALIZED PRODUCT RULE:

If some event can occur in n_1 different ways, and if, following this event, a second event can occur in n_2 different ways, and following this second event, a third event can occur in n_3 different ways, ..., then the number of ways all the events can occur in the order indicated is $n_1 \cdot n_2 \cdot n_3 \cdot \dots$

PRODUCT RULE IN TERMS OF SETS:

If $A_1, A_2, \dots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements in each set. If $n(A_i)$ denotes the number of elements in set A_i , then $n(A_1 \times A_2 \times \dots \times A_m) = n(A_1) \cdot n(A_2) \cdot \dots \cdot n(A_m)$

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Lecture No.30 Permutations important for quiz

FACTORIAL OF A POSITIVE INTEGER:

For each positive integer n , its factorial is defined to be the product of all the integers from 1 to n and is denoted $n!$. Thus $n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$

In addition, we define $0! = 1$

REMARK:

$n!$ Can be recursively defined as

Base: $0! = 1$

Recursion $n! = n(n-1)!$ For each positive integer n .

COUNTING FORMULAS:

From a given set of n distinct elements, one can choose k elements in different ways. The number of selections of elements varies according to as:

- Elements may or may not be repeated.
- The order of elements may or may not matter.

How many conditions of COUNTING FORMULAS?

These two conditions therefore lead us to four counting methods summarized in the following table.

	ORDER MATTERS	ORDER DOES NOT MATTER
REPETITION ALLOWED	k-sample	k-selection
REPETITION NOT ALLOWED	k-permutation	k-combination

K-SAMPLE:

A k -sample of a set of n elements is a choice of k elements taken from the set of n elements such that the order of elements matters and elements can be repeated.

With k -sample, repetition of elements is allowed, therefore, k need not be less than or equal to n . i.e. k is independent of n .

$$n \cdot n \cdot n \cdots n \text{ (k-times)}$$

$$\text{K-SAMPLE: } = n^k$$

K-PERMUTATION:

A k -permutation of a set of n elements is a selection of k elements taken from the set of n elements such that the order of elements matters but repetition of the elements is not allowed. The number of

k -permutations of a set of n elements is denoted $P(n, k)$ or ${}_n P_k$.

- With k -permutation, repetition of elements is not allowed, therefore $k \leq n$
- All n elements are to be permuted, so that $k = n$.

$$\text{K-PERMUTATION: } \frac{n!}{(n-k)!}$$

Lecture No.31 Combinations

K-COMBINATIONS:

A k-combination of a set of n elements is a choice of k elements taken from the set of n elements such that the order of the elements does not matter and elements can't be repeated.

The symbol $C(n, k)$ denotes the number of k-combinations that can be chosen from a set of n elements.

K-selections are also called k-combinations with repetition allowed or multi sets of size k. K-combinations are also written nC_k . k must be less than or equal to n. $k \leq n$

K-COMBINATIONS: $C(k + n - 1) = {}^{k+n-1}C_k$

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Lecture No.32 K-Combinations

K-SELECTIONS:

A k-selection of a set of n elements is a choice of k elements taken from a set of n elements such that the order of elements does not matter and elements can be repeated.

- K-selections are also called k-combinations with repetition allowed or multi sets of size k.
- With k-selections of a set of n elements repetition of elements is allowed. Therefore k need not to be less than or equal to n.

K-COMBINATIONS: $C(k + n - 1) = {}^{k+n-1}C_k$

ORDERED AND UNORDERED PARTITIONS:

	ORDER MATTERS	ORDER DOES NOT MATTER
REPETITION ALLOWED	k-sample	k-selection
REPETITION NOT ALLOWED	k-permutation	k-combination

An unordered partition of a finite set S is a collection $[A_1, A_2, \dots, A_k]$ of disjoint (nonempty) subsets of S (called cells) whose union is S . The partition is ordered if the order of the cells in the list counts.

Let S contain n elements and let $n_1, n_2, \dots, n_k$ be positive integers with $n_1 + n_2 + \dots + n_k = n$.

Then there exist $S = \frac{n!}{n_1! n_2! n_3! \dots n_k!}$ is often called a multinomial coefficient, and is denoted by the symbol.

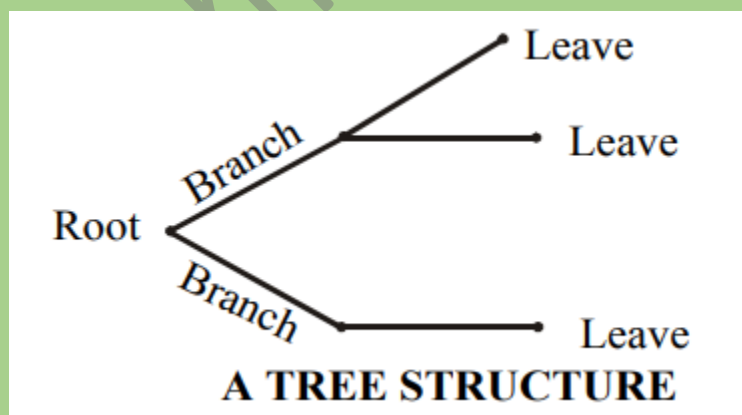
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Lecture No.33 Tree Diagram

TREE DIAGRAM:

A tree diagram is a useful tool to list all the logical possibilities of a sequence of events where each event can occur in a finite number of ways.

A tree is normally constructed from left to right.



THE INCLUSION-EXCLUSION PRINCIPLE:

1. If A and B are disjoint finite sets, then
 $n(A \cup B) = n(A) + n(B)$ the is sum rule
2. If A and B are finite sets, then
 $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ Thus we subtract

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Lecture No.34 Inclusion-Exclusion Principle

INCLUSION-EXCLUSION PRINCIPLE FOR 3 AND 4 SETS:

If A, B, C and D are finite sets, then

- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
- $n(A \cup B \cup C \cup D) = n(A) + n(B) + n(C) + n(D) - n(A \cap B) - n(A \cap C) - n(A \cap D) - n(B \cap C) - n(B \cap D) - n(C \cap D) + n(A \cap B \cap C) + n(A \cap B \cap D) + n(A \cap C \cap D) + n(B \cap C \cap D) - n(A \cap B \cap C \cap D)$

PIGEONHOLE PRINCIPLE

A function from a set of $k + 1$ or more elements to a set of k elements must have at least two elements in the domain that have the same image in the co-domain. If $k + 1$ or more pigeons fly into k pigeonholes then at least one pigeonhole must contain two or more pigeons.

GENERALIZED PIGEONHOLE PRINCIPLE:

A function from a set of $n \cdot k + 1$ or more elements to a set of n elements must have at least $k + 1$ element in the domains that have the same image in the co-domain. If $n \cdot k + 1$ or more pigeons fly into n pigeonholes then at least one pigeonhole must contain $k + 1$ or more pigeons.

PIGEONHOLE PRINCIPLE:

If N pigeons fly into k pigeonholes then at least one pigeonhole must contain (N/k) or more pigeons.

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Lecture No.35 Probability IMP for QUIZ

Probability

DEFINITION:

1. An experiment is a procedure that yields a given set of possible outcomes. The sample space of the experiment is the set of possible outcomes. An event is a subset of the sample space.
2. Let S be a finite sample space such that all the outcomes are equally likely to occur. Its probability of an event E , which is a subset of sample space S , is

$$p(E) = \frac{\text{the number of outcomes in } E}{\text{the number of total outcomes in } S} = \frac{n(E)}{n(S)}$$

Since $\phi \subseteq E \subseteq S$ therefore, $0 \leq n(E) \leq n(S)$. It follows that the probability of an event is always between 0 and 1.

3. Let E be an event in a sample space S , the complement of E is the event that occurs if E does not occur. It is denoted by E^c . Note that $E^c = S \setminus E$

EXAMPLE:

Let E be the event that an even number occurs when a die is tossed. Then E^c is the event that an odd number occurs.

Let E be an event in a sample space S . The probability of the complementary event E^c of E is given by $P(E^c) = 1 - P(E)$.

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Lecture No.36

Laws of Probability

Laws of Probability important

If A and B are two disjoint (mutually exclusive) events of a sample space S , then

$$P(A \cup B) = P(A) + P(B)$$

In words, the probability of the happening of an event A or an event B or both is equal to the sum of the probabilities of event A and event B provided the events have nothing in common.

PROOF:

By inclusion - exclusion principle for mutually disjoint sets,

$$n(A \cup B) = n(A) + n(B)$$

Dividing both sides by $n(S)$, we get

$$\frac{n(A \cup B)}{n(S)} = \frac{n(A) + n(B)}{n(S)}$$

$$\frac{n(A \cup B)}{n(S)} = \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)}$$

$$\Rightarrow P(A \cup B) = P(A) + P(B)$$

GENERAL ADDITION LAW OF PROBABILITY

If A and B are any two events of a sample space S,

Then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

PROOF:

The event $A \cup B$ may be written as the union of two disjoint events $A \setminus B$ and B. i.e., $A \cup B = (A \setminus B) \cup B$. Hence, by addition law of probability (for disjoint events)

$$P(A \cup B) = P(A \setminus B) + P(B)$$

$$P(A \cup B) = [P(A) - P(A \cap B)] + P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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Lecture# 37 Conditional probability

CONDITIONAL PROBABILITY

Let E and F be two events in the sample space of an experiment with $P(F) \neq 0$. The conditional probability of E given F, denoted by $P(E|F)$, is defined as

$$\text{CONDITIONAL PROBABILITY} = P(E|F) = \frac{P(E \cap F)}{P(F)}$$

MULTIPLICATION THEOREM

Let E and F be two events in the sample space of an experiment, then

$$P(E \cap F) = P(F) P(E|F)$$

$$P(E \cap F) = P(E) P(F|E)$$

Let $E_1, E_2, \dots, E_n$ be events in the sample space of an experiment, then $P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1) P(E_2|E_1) P(E_3|E_1 \cap E_2) \dots P(E_n|E_1 \cap E_2 \cap \dots \cap E_{n-1})$.

INDEPENDENCE:

An event A is said to be independent of an event B if the probability that A occurs is not influenced by whether B has or has not occurred. That is, $P(A|B) = P(A)$.

It follows then from the Multiplication Theorem that,

$$P(A \cap B) = P(B) P(A|B) = P(B) P(A)$$

We also know that,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$\therefore$ Because $P(A \cap B) = P(A)P(B)$, due to independence

$$P(B|A) = \frac{P(A)P(B)}{P(A)}$$

$$P(B|A) = P(B)$$

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Lecture# 38 Random variable

Random variable

A random variable X is a rule that assigns a numerical value to each outcome in a sample Space S.

OR

It is a function which maps each outcome of the sample space into the set of real numbers.

- A random variable is also called a chance variable, or a stochastic variable (not called simply a variable, because it is a function).
- Random variables are usually denoted by capital letters such as X, Y, Z; and the values taken by them are represented by the corresponding small letters.

PROBABILITY DISTRIBUTION OF A RANDOM VARIABLE:

Let $X(S) = \{x_1, x_2, \dots, x_n\}$ be the range space of a random variable X defined on a finite sample space S. Define a function f on X(S) as follows:

$$F(x_i) = P(X = x_i)$$

= sum of probabilities of points in S whose image is x_i . This function f is called the **probability distribution** or the probability function of X. The probability distribution f of X is usually given in the form of a table.

x_1	x_2	$x_{\dots\dots\dots}$	x_n
$F(x_1)$	$F(x_2)$	$F(x_{\dots\dots\dots})$	$F(x_n)$

The distribution f satisfies the conditions.

Important for Quiz and formulae

- I. $f(x_i) \geq 0$
- II. $\sum_{i=1}^n f(x_i) = 1$

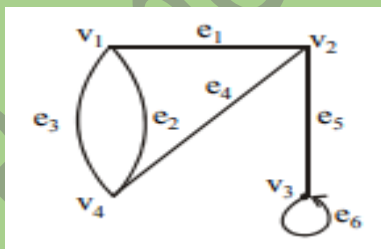
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Lecture# 39 Introduction to Graphs

GRAPH:

A graph is a non-empty set of points called vertices and a set of line segments joining pairs of vertices called edges. Formally, a graph G consists of two finite sets:

- A set $V=V(G)$ of vertices (or points or nodes)
- A set $E=E(G)$ of edges; where each edge corresponds to a pair of vertices.



The graph G with

$$V(G) = \{v_1, v_2, v_3, v_4, v_5\} \text{ and}$$

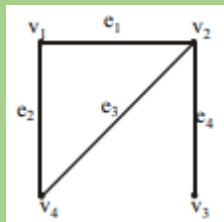
$$E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6\}$$

- I. An edge connects either one or two **vertices called its endpoints** (edge e_1 connects vertices v_1 and v_2 described as $\{v_1, v_2\}$ i.e v_1 and v_2 are the endpoints of an edge e_1).
- II. An edge with **just one endpoint is called a loop**. Thus a loop is an edge that connects a vertex to itself (e.g., edge e_6 makes a loop as it has only one endpoint v_3).
- III. Two vertices that are connected **by an edge are called adjacent**; and a vertex that is an endpoint of a loop is said to be adjacent to it.
- IV. An edge is said to be **incident on each of its endpoints** (i.e. e_1 is incident on v_1 and v_2).
- V. A vertex on which no edges **are incident is called isolated** (e.g., v_5).
- VI. Two distinct edges with the same set of end **points are said to be parallel** (i.e. e_2 & e_3).

SIMPLE GRAPH

A simple graph is a graph that does not have any loop or parallel edges.

EXAMPLE:



$$V(G) = \{v_1, v_2, v_3, v_4, v_5\} \text{ \& \; } E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6\}$$

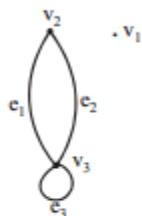
DEGREE OF A VERTEX:

Let G be a graph and v a vertex of G . The degree of v , denoted $\deg(v)$, equals the number of edges that are incident on v , with an edge that is a loop counted twice.

- The total degree of G is the sum of the degrees of all the vertices of G .
- The degree of a loop is counted twice.

EXAMPLE:

For the graph shown



$\deg(v_1) = 0$, since v_1 is isolated vertex.

$\deg(v_2) = 2$, since v_2 is incident on e_1 and e_2 .

$\deg(v_3) = 4$, since v_3 is incident on e_1, e_2 and the loop e_3 .

Total degree of $G = \deg(v_1) + \deg(v_2) + \deg(v_3)$

$$= 0 + 2 + 4$$

$$= 6$$

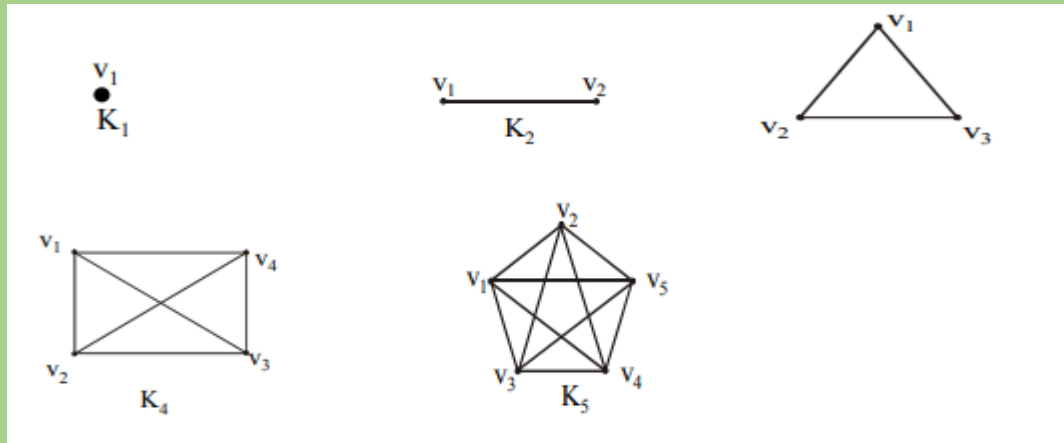


THE HANDSHAKING THEOREM

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_n)$$

COMPLETE GRAPH:

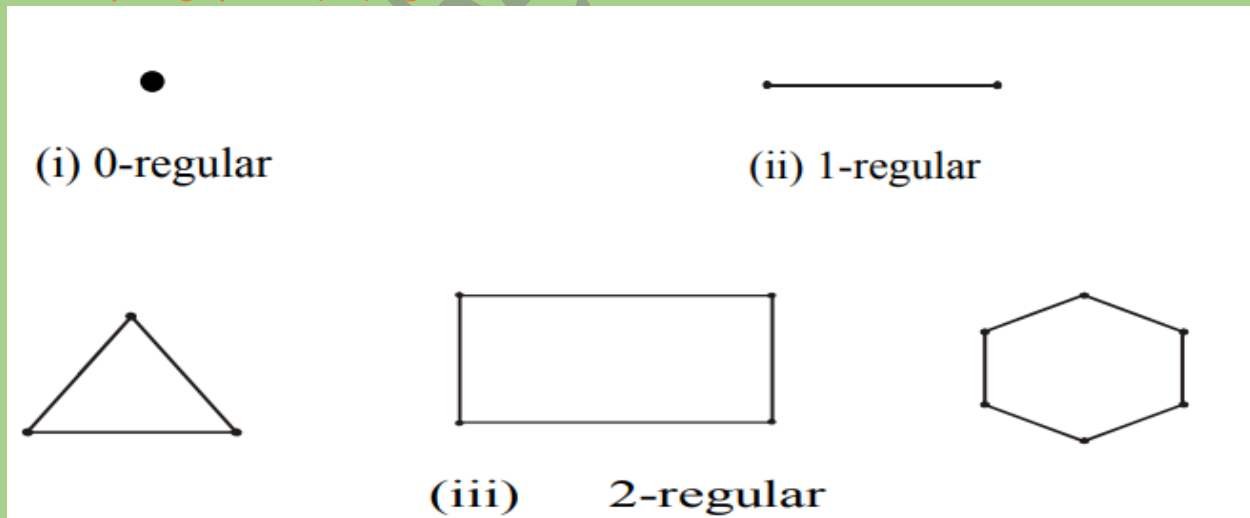
A complete graph on n vertices is a simple graph in which each vertex is connected to every other vertex and is denoted by K_n (K_n means that there are n vertices). The following are complete graphs K_1 , K_2 , K_3 , K_4 and K_5 .



REGULAR GRAPH:

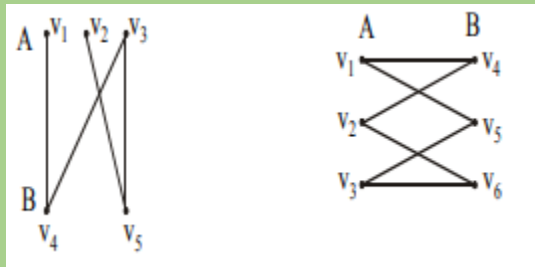
A graph G is regular of degree k or k -regular if every vertex of G has degree k . In other words, a graph is regular if every vertex has the same degree. Following are some regular graphs.

The complete graph K_n is $(n-1)$ regular.



BIPARTITE GRAPH:

A bipartite graph G is a simple graph whose vertex set can be partitioned into two mutually disjoint non empty subsets A and B such that the vertices in A may be connected to vertices in B , but no vertices in A are connected to vertices in A and no vertices in B are connected to vertices in B .

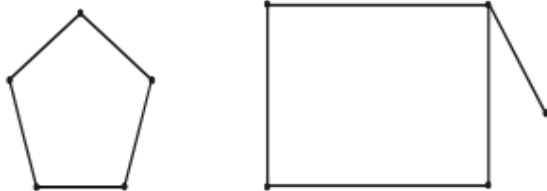


DETERMINING BIPARTITE GRAPHS:

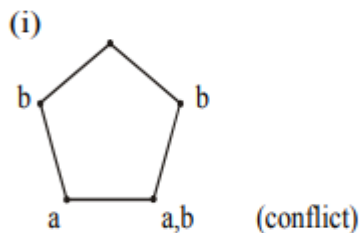
The following labeling procedure determines whether a graph is bipartite or not. 1. Label any vertex a 2. Label all vertices adjacent to a with the label b. 3. Label all vertices that are adjacent to a vertex just labeled b with label a. 4. Repeat steps 2 and 3 until all vertices got a distinct label (a bipartite graph) or there is a conflict i.e., a vertex is labeled with a and b (not a bipartite graph).

EXERCISE:

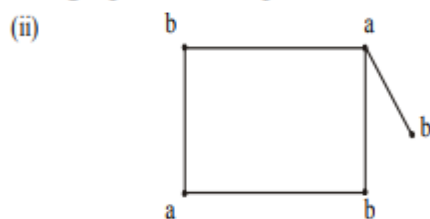
Find which of the following graphs are bipartite. Redraw the bipartite graph so that its bipartite nature is evident.



Answers



The graph is not bipartite.



COMPLETE BIPARTITE GRAPH:

A complete bipartite graph on $(m+n)$ vertices denoted $K_{m,n}$ is a simple graph whose vertex set can be partitioned into two mutually disjoint non empty subsets A and B containing m and n vertices respectively, such that each vertex in set A is connected (adjacent) to every vertex in set B , but the



vertices within a set are not connected.

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Lecture# 40 Paths and Circuits

IMP DEFINITIONS

PATHS AND CIRCUITS

Path is a route along edges that start at a vertex and end at a vertex. Circuit is a path that begins and ends at the same vertex. A graph is connected if for any two vertices there at least one path connecting them. Otherwise graph is disconnected.

Equivalently:

Let G be a graph and let v and w be vertices in graph G .

1. WALK

A walk from v to w is a finite alternating sequence of adjacent vertices and edges of G .

Thus a walk has the form

$$v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$$

Where the v 's represent vertices, the e 's represent edges $v_0=v$, $v_n=w$, and for all $i = 1, 2 \dots n$, v_{i-1} and v_i are endpoints of e_i .

The trivial walk from v to v consists of the single vertex v .

2. CLOSEDWALK

A closed walk is a walk that starts and ends at the same vertex.

3. CIRCUIT

A circuit is a closed walk that does not contain a repeated edge. Thus a circuit is a walk of the form $v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$ where $v_0 = v_n$ and all the e_i s are distinct.

4. SIMPLE CIRCUIT

A simple circuit is a circuit that does not have any other repeated vertex except the first and last. Thus a simple circuit is a walk of the form $v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n$ where all the e_i s are distinct and all the v_j s are distinct except that $v_0 = v_n$

5. PATH

A path from v to w is a walk from v to w that does not contain a repeated edge. Thus a path from v to w is a walk of the form $v = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n = w$ where all the e_i s are distinct (that is $e_i \neq e_k$ for any $i \neq k$).

6. SIMPLE PATH

A simple path from v to w is a path that does not contain a repeated vertex. Thus a simple path is a walk of the form $v = v_0 e_1 v_1 e_2 \dots v_{n-1} e_n v_n = w$ where all the e_i s are distinct and all the v_j s are also distinct (that is, $v_j \neq v_m$ for any $j \neq m$).

SUMMARY

	Repeated Edge	Repeated Vertex	Starts and Ends at Same Point
walk	allowed	Allowed	allowed
closed walk	allowed	Allowed	yes(means, where it starts also ends at that point)
circuit	no	Allowed	yes
simple circuit	no	first and last only	yes
path	no	Allowed	allowed
simple path	no	no	No

EULER CIRCUITS

Let G be a graph. An Euler circuit for G is a circuit that contains every vertex and every edge of G . That is, an Euler circuit for G is sequence of adjacent vertices and edges in G that starts and ends at the same vertex uses every vertex of G at least once, and used every edge of G exactly once.

EULER PATH DEFINITION:

Let G be a graph and let v and w be two vertices of G . An Euler path from v to w is a sequence of adjacent edges and vertices that starts at v , ends at w , passes through every vertex of G at least once, and traverses every edge of G exactly once.

COROLLARY

Let G be a graph and let v and w be two vertices of G . There is an Euler path from v to w if, and only if, G is connected, v and w have odd degree and all other vertices of G have even degree.

HAMILTONIAN CIRCUITS

Given a graph G , a Hamiltonian circuit for G is a simple circuit that includes every vertex of G . That is, a Hamiltonian circuit for G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once.

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Lecture# 41 Matrix Representations of Graphs

IMP DEFINITIONS

MATRIX:

An $m \times n$ matrix A over a set S is a rectangular array of elements of S arranged into m rows and n columns:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

SQUARE MATRIX:

A matrix for which the number of rows and columns are equal is called a square matrix. A square matrix A with m rows and n columns (size $m \times n$) but $m=n$ (i.e. of order $n \times n$) has the form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1i} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2i} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ii} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{ni} & \cdots & a_{nn} \end{bmatrix}$$

Diagonal entries

TRANPOSE OF A MATRIX:

The transpose of a matrix A of size $m \times n$, is the matrix denoted by A^t of size $n \times m$, obtained by writing the rows of A , in order, as columns. (Or we can say that transpose of a matrix means “write the rows instead of columns or write the columns instead of rows”).

Thus if

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \text{then } A^t = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

SYMMETRIC MATRIX:

A square matrix $A = [a_{ij}]$ of size $n \times n$ is called symmetric if, and only if, $A^t = A$ i.e., for all $i, j = 1, 2, \dots, n$, $a_{ij} = a_{ji}$

MATRIX MULTIPLICATION:

Suppose A and B are two matrices such that the number of columns of A is equal to the number of rows of B , say A is an $m \times p$ matrix and B is a $p \times n$ matrix. Then the product of A and B , written AB , is the $m \times n$ matrix whose ij th entry is obtained by multiplying the elements of the i th row of A by the corresponding elements of the j th column of B and then adding;

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ip} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & & \vdots & & \vdots \\ b_{p1} & \cdots & b_{pj} & \cdots & b_{pn} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1n} \\ \vdots & & \vdots & & \vdots \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{in} \\ \vdots & & \vdots & & \vdots \\ c_{m1} & \cdots & c_{mj} & \cdots & c_{mn} \end{bmatrix}$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}$$

If the number of columns of A is not equal to the number of rows of B, then the product AB is not defined.

ADJACENCY MATRIX OF A GRAPH:

Let G be a graph with ordered vertices $v_1, v_2, \dots, v_n$. The adjacency matrix of G is the matrix $A = [a_{ij}]$ over the set of non-negative integers such that a_{ij} = the number of edges connecting v_i and v_j for all $i, j = 1, 2, \dots, n$.

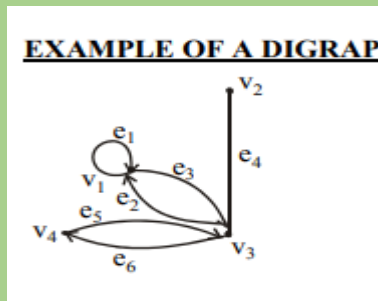
The adjacency matrix say $A = [a_{ij}]$ is also defined as

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

DIRECTED GRAPH:

A directed graph or digraph, consists of two finite sets: a set $V(G)$ of vertices and a set $D(G)$ of directed edges, where each edge is associated with an ordered pair of vertices called its end points.

EXAMPLE OF A DIGRAP

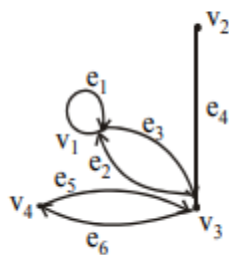


ADJACENCY MATRIX OF A DIRECTED GRAPH:

Let G be a graph with ordered vertices $v_1, v_2, \dots, v_n$. The adjacency matrix of G is the matrix $A = [a_{ij}]$ over the set of non-negative integers such that a_{ij} = the number of arrows from v_i to v_j for all $i, j = 1, 2, \dots, n$.

EXAMPLE:

A directed graph with its adjacency matrix is shown



$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

is the adjacency matrix

INCIDENCE MATRIX OF A SIMPLE GRAPH:

Let G be a graph with vertices $v_1, v_2, \dots, v_n$ and edges $e_1, e_2, \dots, e_m$. The incidence matrix of G is the matrix $M = [m_{ij}]$ of size $n \times m$ defined by

$$m_{ij} = \begin{cases} 1 & \text{if the vertex is incident on the edge } e_j \\ 0 & \text{otherwise} \end{cases}$$

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Lecture # 42

Isomorphism of graphs IMP

DEFINITIONS

ISOMORPHISM OF GRAPHS

A property P is called an isomorphic invariant if, and only if, given any graphs G and G' ; if G has property P and G' is isomorphic to G , then G' has property P .

THEOREM OF ISOMORPHIC INVARIANT:

Each of the following properties is an invariant for graph isomorphism, where n , m and k are all non-negative integers, if the graph:

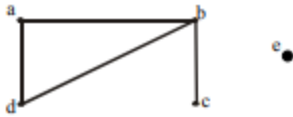
- I. Has an n vertex.
- II. Has an m edge.
- III. Has a vertex of degree k .
- IV. Have m vertices of degree k .
- V. Has a circuit of length k .
- VI. Has a simple circuit of length k .
- VII. Have m simple circuits of length k .
- VIII. Is connected.
- IX. Has an Euler circuit.
- X. Has a Hamiltonian circuit.

DEGREE SEQUENCE:

The degree sequence of a graph is the list of the degrees of its vertices in non-increasing order.

EXAMPLE:

Find the degree sequence of the following graph.

**SOLUTION:**

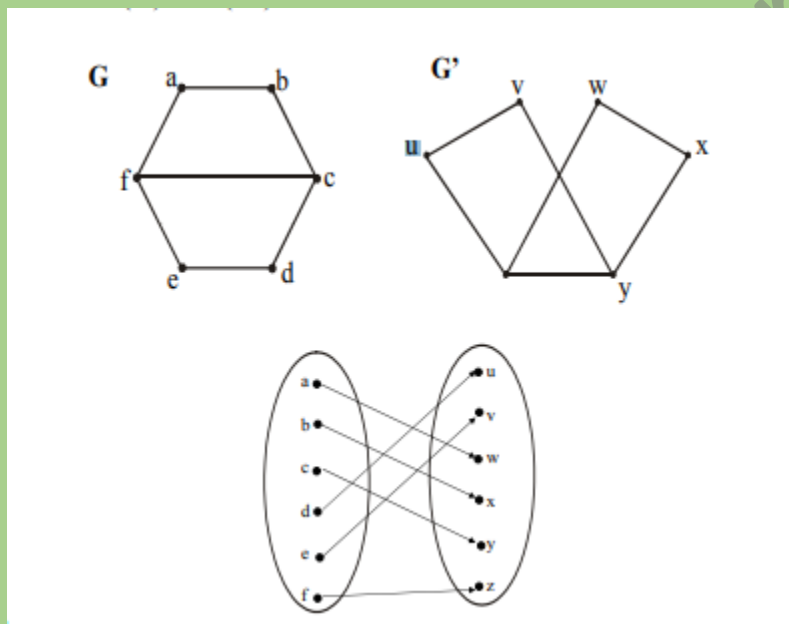
Degree of $a = 2$, Degree of $b = 3$, Degree of $c = 1$,
Degree of $d = 2$, Degree of $e = 0$

By definition, degree of the vertices of a given graph should be in decreasing (non-increasing) order.

Therefore Degree sequence is: 3, 2, 2, 1, 0

ALTERNATIVE SOLUTION:

We shall prove that the graphs G and G' are isomorphic. Define a function $f: V(G) \rightarrow V(G')$ as follows.

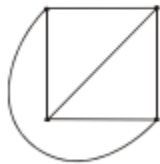
**Lecture# 43 planar graphs****IMP DEFINITIONS****PLANAR GRAPHS GRAPH COLORING****DEFINITION:**

A graph is called planar if it can be drawn in the plane without any edge crossed (crossing means the intersection of lines). Such a drawing is called a plane drawing of the graph.

OR

You can say that a graph is called planar in which the graph crossing number is “0”.

EXAMPLES:

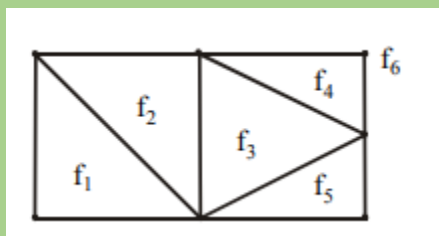


The graphs given above are planar. In the first figure edges are crossed but it can be redrawn in second figure where edges are not crossed, so called planar.

Infinite face and faces

A plane drawing of a planar graph divides the plane into regions, including an unbounded region, called faces.

The unbounded region is called the infinite face.



EULER'S FORMULA

$$f = e - v + 2$$

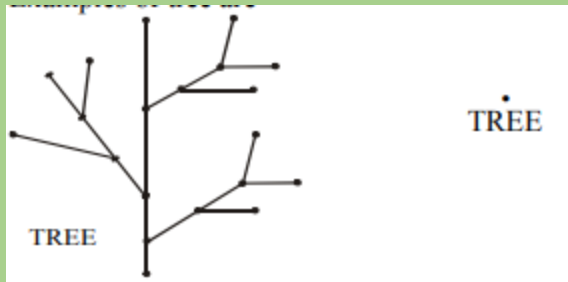
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Lecture# 44 Trees IMP DEFINITIONS

TREE:

A tree is a connected graph that does not contain any non-trivial circuit. (I.e. it is circuit free). A trivial circuit is one that consists of a single vertex.

Examples of tree are



SOME SPECIAL TREES

➤ TRIVIAL TREE:

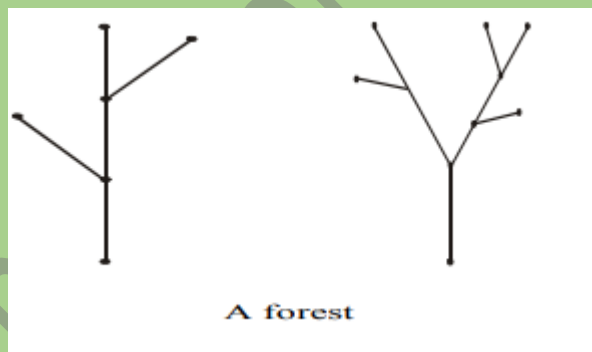
A graph that consists of a single vertex is called a trivial tree or degenerate tree.

➤ EMPTY TREE

A tree that does not have any vertices or edges is called an empty tree.

➤ FOREST

A graph is called a forest if, and only if, it is circuit-free. OR “Any non-connected graph that contains no circuit is called a forest.” Hence, it clears that the connected components of a forest are trees.



As in both the graphs above, there exists no circuit, so called forest.

PROPERTIES OF TREES:

1. A tree with n vertices has $n - 1$ edges (where $n \geq 0$).
2. Any connected graph with n vertices and $n - 1$ edges is a tree.
3. A tree has no non-trivial circuit; but if one new edge (but no new vertex) is added to it, then the resulting graph has exactly one non-trivial circuit.
4. A tree is connected, but if any edge is deleted from it, then the resulting graph is not connected.
5. Any tree that has more than one vertex has at least two vertices of degree 1.

6. A graph is a tree iff there is a unique path between any two of its vertices.

Differences b/t terminal vertex and branch vertex?

A vertex of degree 1 in a tree is called a terminal vertex or a leaf and a vertex of degree greater than 1 in a tree is called an internal vertex or a branch vertex.

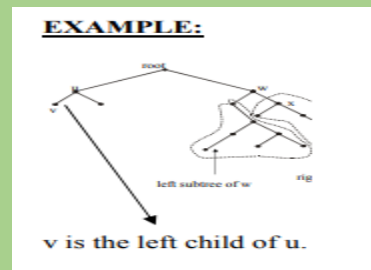
ROOTED TREE:

- A **rooted tree** is a tree in which one vertex is distinguished from the others and is called **the root**.
- **The level of a vertex** is the number of edges along the unique path between it and the root.
- The **height** of a rooted tree is the maximum level to any vertex of the tree.
- The **children** of any internal vertex v are all those vertices that are adjacent to v and are one level farther away from the root than v .
- If w is a child of v , then **v is called the parent of w** .
- Two vertices that are both children of **the same parent are called siblings**.
- Given vertices v and w , if v lies on the unique path between w and the root, then v is an ancestor of w and w is a descendant of v .

BINARY TREE

A **binary tree** is a rooted tree in which every internal vertex has at most two children. Every child in a binary tree is designated either a left child or a right child (but not both).

A **full binary tree** is a binary tree in which each internal vertex has exactly two children.

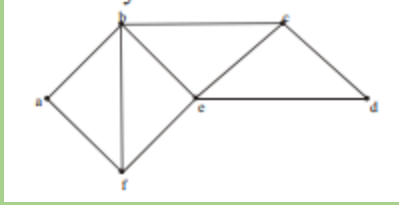


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Lecture # 45 Spanning Trees

SPANNING TREES:

Suppose it is required to develop a system of roads between six major cities. A survey of the area revealed that only the roads shown in the graph could be constructed.



SPANNING TREE:

A spanning tree for a graph G is a sub graph of G that contains every vertex of G and is a tree.

- I. Every connected graph has a spanning tree.
- II. A graph may have more than one spanning trees.
- III. Any two spanning trees for a graph have the same number of edges.
- IV. If a graph is a tree, then its only spanning tree is itself.

KIRCHHOFF'S THEOREM OR MATRIX - TREE THEOREM

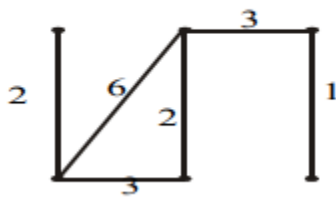
Let M be the matrix obtained from the adjacency matrix of a connected graph G by changing all 1's to -1's and replacing each diagonal 0 by the degree of the corresponding vertex. Then the number of spanning trees of G is equal to the value of any cofactor of M .

WEIGHTED GRAPH

A WEIGHTED GRAPH is a graph for which each edge has an associated real number weight. The sum of the weights of all the edges is the total weight of the graph.

EXAMPLE:

The figure shows a weighted graph



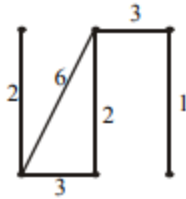
with total weight is $2 + 6 + 3 + 2 + 3 + 1 = 17$

MINIMAL SPANNING TREE:

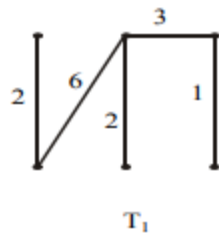
A minimal spanning tree for a weighted graph is a spanning tree that has the least possible total weight compared to all other spanning trees of the graph. If G is a weighted graph and e is an edge of G then $w(e)$ denotes the weight of e and $w(G)$ denotes the total weight of G .

EXERCISE:

Find the three spanning trees of the weighted graph below. Also indicate the minimal spanning tree.



SOLUTION:



T_1

$$w(T_1)=14$$

Pray for all helpers 🙏🙏👍👍

Thank you 🙏🙏👍👍,

Best of luck